

Ex: Find MED between

$X = 'VED'$ $Y = 'DEV'$

T	j=0	D	E	V
i=0	0	1	2	3
V	1	2	3	2
E	2	3	2	3
D	3	2	3	4

$$T[1][1] = \min \{ T[0][1] + 1, T[0][0] + k, T[1][0] + 1 \} \quad (k=2)$$

$$= \min \{ 1 + 1, 0 + 2, 1 + 1 \}$$

$$= 2$$

$$T[1][2] = \min \{ T[0][2] + 1, T[0][1] + k, T[1][1] + 1 \} \quad (k=2)$$

$$= \min \{ 2 + 1, 1 + 2, 2 + 1 \}$$

$$= 3$$

$$T[1][3] = \min \{ T[0][3] + 1, T[0][2] + 0, T[1][2] + 1 \} \quad (k=0)$$

$$= \min \{ 3 + 1, 2 + 0, 3 + 1 \}$$

$$= 2$$

$$T[2][1] = \min \{ T[1][1] + 1, T[1][0] + k, T[2][0] + 1 \}$$

Here $k=2$

$$\therefore T[2][1] = \text{Min} \{ 2+1, 1+2, 2+1 \} = 3$$

$$\rightarrow T[2][2] = \text{Min} \left\{ \begin{array}{l} T[1][2] + 1, \\ T[1][1] + k, \\ T[2][1] + 1 \end{array} \right\}$$

$$= \text{Min} \{ 3+1, 2+0, 3+1 \}$$

$$= 2$$

Here $k=0$

because $x[2] = y[2]$

$$\rightarrow T[2][3] = \text{Min} \left\{ \begin{array}{l} T[1][3] + 1, \\ T[1][2] + k, \\ T[2][2] + 1 \end{array} \right\}$$

$$= \text{Min} \{ 2+1, 3+2, 2+1 \} \quad (k=2)$$

$$= 3$$

$$\rightarrow T[3][1] = \text{Min} \left\{ \begin{array}{l} T[2][1] + 1, \\ T[2][0] + k, \\ T[3][0] + 1 \end{array} \right\}$$

$$= \text{Min} \{ 3+1, 2+0, 3+1 \}$$

$$= 2$$

$k=0$

$$\rightarrow T[3][2] = \text{Min} \left\{ \begin{array}{l} T[2][2] + 1, \\ T[2][1] + k, \\ T[3][1] + 1 \end{array} \right\}$$

$k=2$
 $x[3] \neq y[2]$

$$= \text{Min} \{ 2+1, 3+2, 2+1 \}$$

$$= 3$$

$$\rightarrow T[3][3] = \text{Min} \left\{ \begin{array}{l} T[2][3] + 1, \\ T[2][2] + k, \\ T[3][2] + 1 \end{array} \right\} \quad k=2$$

$$= \text{Min} \left\{ \begin{array}{l} 3+1, \\ 2+2, \\ 3+1 \end{array} \right\}$$

$$= 4$$

To Convert VED $\rightarrow$ DEV.

$$\boxed{MED=4}$$